UTSAV PIPERSANIYA

AI Engineer — LLM Applications, RAG, Python
Jabalpur, Madhya Pradesh, India | +91 7477011067 | upipersaniya@gmail.com
LinkedIn: linkedin.com/in/utsavpipersaniya | GitHub: github.com/Utsav-x | Live: drops.raritaclub.com
Available immediately | Open to remote (India), or hybrid/on-site in Pune, Indore, Gurugram

SUMMARY

Engineer who ships **LLM systems into production**, not demos. Built and operates an enterprise AI assistant using retrieval-augmented generation over **pgvector** with **OpenAI** models, which also queries live platform data so answers are grounded in the customer's real records rather than documents alone. As a freelance engineer, also built a provider-agnostic AI layer — vision extraction, search-grounded valuation, and a knowledge-base support agent with structured output and automatic human escalation. Backed by nearly 3 years of production Python and PostgreSQL engineering, which is what makes the retrieval, latency, cost and failure-handling around a model work rather than the prompt alone.

TECHNICAL SKILLS

- **LLM & AI** — Retrieval-augmented generation (RAG), embeddings, vector search, pgvector, semantic retrieval, prompt engineering, system prompts, structured JSON output, schema-constrained generation, tool / function calling over live data, multimodal vision extraction, search grounding, context-window management, human-in-the-loop escalation, LLM cost and quota control, retry and backoff, provider abstraction
- **Model APIs** — OpenAI API, Google Gemini API, Anthropic Claude API
- **Languages** — Python, SQL, JavaScript, TypeScript
- **Data** — PostgreSQL, pgvector, SQL query optimization, ETL pipelines, data cleaning and normalization, entity resolution, pandas, NumPy
- **Backend** — Django, Django REST Framework, FastAPI, Celery, Celery Beat, Redis, REST APIs, WebSockets, async processing
- **Infrastructure** — Docker, Linux, Nginx, Gunicorn, Google Cloud Platform, Azure Blob Storage, Git, Sentry

EXPERIENCE

Backend / AI Engineer | Sustainext | Dec 2023 – Jul 2026

Enterprise ESG and sustainability reporting platform.

- Built **Sustainext Chat**, a production AI assistant that answers both general sustainability questions and questions about the customer's own data held in the platform.
- Implemented **retrieval-augmented generation over pgvector** — embeddings stored and searched inside PostgreSQL, with retrieval over the sustainability knowledge corpus feeding an **OpenAI** generation step.
- Went beyond document RAG: the agent also **queries live platform data**, so responses reflect the customer's actual ESG records rather than a static corpus. Non-technical users get natural-language access to reporting data.
- Wrote the orchestration directly against the provider SDK — prompt assembly, retrieval, context budgeting and response handling — rather than adopting a framework, which kept latency and token cost under direct control.
- Built the surrounding production concerns that decide whether an AI feature survives contact with users: caching, asynchronous execution off the request thread via Celery, and failure handling that degrades rather than breaks.
- Built and owned a fully automated ETL pipeline that ingests clients' third-party data into PostgreSQL on a Celery Beat schedule, with the transform stage in **pandas and NumPy**, with live progress streamed over SSE; it made that data onboardable and **brought in 5 new clients**.
- Built an in-house emission-factor database that **reduced third-party API costs**.
- Also owned the platform's non-AI backend: 100+ REST endpoints, the emissions analysis engine, and the Redis caching layer.
- **Cut API response times ~50%** through PostgreSQL indexing and ORM tuning, and **cut long-running workload times ~60%** by moving processing onto Celery and Redis — both measured before and after.

Freelance Backend Engineer | Rarita Technologies, Gurugram | Jul 2026 – Present

Collectibles marketplace and live-auction platform. Python, Django, Gemini, PostgreSQL, Redis, Celery.

Provider-Agnostic AI Vision & Valuation Pipeline

- Built a **two-stage LLM pipeline** for collectible identification: Stage 1 does deterministic multimodal extraction from user photos into a structured catalogue card; Stage 2 runs a separate **web-search-grounded** valuation against sold comparables. Splitting them stopped price speculation leaking into identification.
- Designed a **provider abstraction** selected by configuration — Gemini implemented, OpenAI and Claude behind the same interface — so the model vendor is a setting, not a rewrite.
- Enforced **structured JSON output** with parsing, sanitization and price validation, plus retry with backoff, so malformed or hallucinated model output fails closed instead of reaching the database.
- Cut inference cost with a **valuation cache** keyed on item identity: exact-key lookup first, then a PostgreSQL `TrigramSimilarity` fuzzy fallback, so two users scanning the same item share one expensive search-grounded call. Added a per-user daily quota with local-midnight rollover.

Knowledge-Base Support Agent

- Built an **AI first responder** on support conversations, running off the request thread via Celery.
- Grounded it strictly on a cached knowledge base assembled from the FAQ table and legal documents, with a bounded prompt and bounded conversation history to keep token cost predictable.
- Made the model return a structured `{answer, escalate}` decision and **route to a human automatically** for anything account-specific, financial, or sensitive — the safety boundary is enforced in the schema, not just requested in the prompt.

Rarità — Real-Time Auction Marketplace

Django 5, DRF, Channels, Celery, PostgreSQL, Redis, Docker, GCP.

- Built a **Redis-authoritative bidding engine**: authoritative state in a Redis hash, a single atomic Lua script for the accept path, integer-paise arithmetic, monotonic sequence numbers, and idempotent batched persistence to PostgreSQL that survives a crash mid-write.
- Built a time-decayed trending ranking on hourly Redis hashes with fixed recompute cost and graceful degradation on Redis failure.

RaritaDrops — live at drops.raritaclub.com

- Shipped a real-time auction platform solo — architecture through deployment and ongoing operation. Django + Channels backend, Next.js 15 / React 19 frontend.

EDUCATION

B.Tech — Gyan Ganga Institute of Technology and Sciences, 2023 | CGPA 8.34

CERTIFICATIONS

- **Technology Virtual Experience Program** — Deloitte via Forage, 2023
- **Data Analytics Virtual Internship** — Trainity, 2023